



Ophir LMMMeasurement COM Object

User Manual

OphirLMMeasurement COM Object

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1 Revision History

15	20-Apr-26	1. Rewrote Data Streams section
14	18-Dec-24	1. Support for connecting to Ariel wirelessly
13	31-Jul-22	1. Mention Visual C++ redistributable
12	05-Jan-21	1. Support for Ariel 2. Support for VS 2019 3. Added GetData Status Codes: Temperature, Alert Hot, Pulse Width, and PfP Energy
11	25-Mar-20	1. Removed XP support 2. Added status to GetData method for reporting pulse frequency 3. Added information about the new interface IColMMMeasurement2 4. Added functions for calibration date 5. Added functions for pulsed power pulse width 6. Added functions for low frequency power pulse frequency
10	11-Apr-19	1. Added Centauri and Juno+ 2. Listed Python and Java as environments we supply a demo for 3. Added Creating an Object and Threading issues sections
09	22-Mar-17	1. Added C++ demo information
08	30-May-16	1. Added StarLite and StarBright 2. Added status report to GetData method for reporting Filter status for sensors that self-detect their Filter State 3. Remove component for 32-bit applications on 64-bit Windows
07	07-Aug-14	1. Added Section Troubleshooting Regsvr32
06	30-Apr-14	1. Added true 64-bit version of OphirLMMeasurement a. Updated Section Additional Registrations b. Added Section Versions of OphirLMMeasurement
05	23-Apr-12	1. For 64-bit Windows the redistributable folder is renamed from x64 to x86_on_x64 2. Updated explanation of using OphirLMMeasurement in 64-bit applications
04	13-Dec-11	1. Added to Section Additional Registrations description of uninstall of drivers as well as silent install/uninstall
03	26-Jul-11	1. Fix status numbers in Section DataReady_Handler . 2. BeamTrack sensor support added to COM object 3. Added section Configuring the Sensor's Measurement Mode 4. Moved list of status codes from section DataReady_Handler to new section GetData Status Codes and expanded the list to include codes returned for BeamTrack sensors 5. Listed MatLab as one of the IDE's that we supply a demo for
02	24-Nov-10	1. Fix Offset Measurement error for new pyroelectric sensors.
01	11-Nov-10	1. Initial Revision

2 Overview

2.1 Introduction

This document describes OphirLMMeasurement. This is the COM object provided by Ophir that allows user applications to interface with all Ophir USB speaking devices (Centauri, StarLite, StarBright, Juno, Juno+, Nova-II, all Pulsar models, USBI, Vega, and Ariel).

Besides their use as standalone, fully featured laser power/energy meters, Ophir devices can also be used through OphirLMMeasurement. This COM object allows system integrators to integrate the measurement expertise of Ophir equipment with in-house analysis packages.

Demo client applications in VB.NET 2010/2019, VC# 2010/2019, VC++ 2010/2019, Python, Java, MatLab, and LabVIEW 8.6 are provided with the StarLab installation and can be found in C:\Program Files\Ophir Optronics\StarLab 4.xx\Automation Examples\Com object.

In addition, OphirLMMeasurement has been tested with Delphi 5. In practice, OphirLMMeasurement can be used in any environment that allows interaction with COM automation servers (although it hasn't been tested with tools other than those mentioned).

Special note for C++ users: Due the intricacy involved in using COM objects in C++, the C++ demo includes a wrapper class which provides a standard C++ interface to the COM object. To use this wrapper class, add OphirLMMeasurement.h and OphirLMMeasurement.cpp to your project, and #include "OphirLMMeasurement.h". The file program.cpp is a demo of how to use the class. See the comments in OphirLMMeasurement.h for more details.

Note: We provide a COM object and not a .NET automation because many clients' applications do not use .NET and it is easy to use a COM object from .NET.

Note: As of StarLab 3.20, the OphirFastX and OphirUsbX ActiveX packages are no longer included with the StarLab installation.

2.2 Improvements in OphirLMMeasurement

2.2.1 Support for All Devices in One Package

Centauri, StarLite, StarBright, Juno, Juno+, Nova-II, Vega, USBI, Pulsar, and Ariel are all supported by OphirLMMeasurement.

OphirLMMeasurement provides a common interface with all of the devices. Therefore devices with similar features should be drop-in compatible in the code.

2.2.2 Console Applications

OphirLMMeasurement doesn't need a window and can therefore work with a console client. This includes Microsoft based applications and other IDE's as well such as MatLab scripts.

2.2.3 Immediate Response

OphirLMMeasurement provides a mode in which each measurement is reported immediately when taken at frequencies of up to 100Hz.

2.2.4 Turbo Mode

Due to improvements in the underlying code, Turbo Mode is no longer necessary when measuring with Pulsar devices.

2.3 USB Details of Devices

The following lists the differences between the various Ophir devices vis-à-vis their USB communications. OphirLMMeasurement hides all this from the client application but the system integrator must be aware of this to ensure he sets up the system with the appropriate hardware support.

• Centauri	Up to 2 channels,	High Speed,	Self-powered,	one interrupt IN endpoint
• Juno	One channel,	Full Speed,	Bus Powered,	up to 5 interrupt IN endpoints
• Juno+	One channel,	Full Speed,	Bus Powered,	up to 5 interrupt IN endpoints
• Nova-II	One channel,	Full Speed,	Self powered,	one interrupt IN endpoint
• Pulsar	Up to 4 channels,	High Speed,	Self powered,	up to 2 interrupt IN endpoints
• USBI	One channel,	Full Speed,	Bus Powered,	one interrupt IN endpoint
• Vega	One channel,	Full Speed,	Self powered,	one interrupt IN endpoint
• StarLite	One channel,	High Speed,	Self powered,	one interrupt IN endpoint
• StarBright	One channel,	High Speed,	Self powered,	one interrupt IN endpoint
• Ariel	One channel,	Full Speed,	Self powered,	one interrupt IN endpoint

3 OphirLMMeasurement: Detailed Description

This section describes registration of OphirLMMeasurement as well as its methods, events, and error codes.

3.1 Registration

StarLab PC software (2.10 and higher) registers OphirLMMeasurement as part of the installation process of the StarLab application. Registration is also possible via the Menu of the StarLab application.

3.1.1 Additional Registrations

After installing the StarLab PC application on one PC, it is possible to register the COM objects on additional PC's without installing the full StarLab package. To do so, use the following steps

1. Install the Visual Studio 2019 Visual C++ redistributable, available at <https://docs.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist>.
2. Copy the necessary files
 - For 32-bit applications (running on 32-bit or 64-bit Windows): From **C:\Program Files\Ophir Optronics\StarLab 3.xx\Automation Examples\Com object\Redistributable\x86** to the target PC
 - For 64-bit applications: From **C:\Program Files\Ophir Optronics\StarLab 3.xx\Automation Examples\Com object\Redistributable\x64** to the target PC.
3. Run **installophircom.bat** file on the target PC. This will install the USB drivers as well as register the OphirLMMeasurement COM object and all other components necessary for full functionality. The following is the contents of **installophircom.bat**
 - rem 1) install COM (this is Ophir specific)
 - regsvr32 OphirLMMeasurement.dll
 - rem 2) install Ophir device drivers (Ophir specific INF files)
 - swapinf StarlabDrivers
4. If there is a need to remove the software from the target PC and to clean its registry, run **RemoveOphirCOM.bat** that is provided. The following is the contents of **RemoveOphirCOM.bat**
 - rem 1) remove Ophir device drivers (Ophir specific INF files)
 - swapinf StarlabDriversRemove
 - rem 2) remove COM (this is Ophir specific)
 - regsvr32 /u OphirLMMeasurement.dll
5. The default COM object installation includes Ophir dialog windows informing the user to disconnect Ophir devices. If it is necessary to hide this message, then the install and uninstall batch files can be run silently. To do this, modify **installophircom.bat** and **RemoveOphirCOM.bat** as follows
 - **installophircom.bat**
 1. regsvr32 /s OphirLMMeasurement.dll
 2. swapinf StarlabDrivers --silent
 - **RemoveOphirCOM.bat**
 1. swapinf StarlabDriversRemove --silent
 2. regsvr32 /u /s OphirLMMeasurement.dll
 - Note: If you chose to install and remove silently, you must verify that the Ophir device has been disconnected before starting the process. Otherwise the installation will fail may leave unwanted information in the OC and you won't know about it

3.1.2 Troubleshooting Regsvr32

Microsoft reports that there are occasions where Regsvr32 doesn't execute properly even though it is being run as an Administrator (see <http://support.microsoft.com/kb/249873>). The way to overcome this is to re-run the Regsvr32 command from an elevated command prompt.

To open an elevated command prompt, follow these steps:

- **Windows 8.1 and Windows 8**

Swipe in from the right edge of the screen, and then tap **Search**. Or, if you are using a mouse, point to the lower-right corner of the screen, and then click **Search**. Type **Command Prompt** in the **Search** box, right-click **Command Prompt**, and then click **Run as administrator**. If you are prompted for an administrator password or for a confirmation, type the password, or click **Allow**.

- **Windows 7 and Windows Vista**

Click **Start**, type **Command Prompt** or **cmd** in the **Search** box, right-click **Command Prompt**, and then click **Run as administrator**. If you are prompted for an administrator password or for a confirmation, type the password, or click **Allow**.

- **Windows 10 and Windows 11**

Click the **Start** menu, type **Command Prompt** or **cmd** in the **Search** box, right-click **Command Prompt**, and then click **Run as administrator**. If you are prompted for an administrator password or for a confirmation, type the password, or click **Allow**.

3.1.3 Versions of OphirLMMeasurement

The OphirLMMeasurement component works in Windows Vista, 7, 8, 8.1, 10, and 11. We provide 2 variations.

- 32-bit component for 32-bit applications.
- 64-bit component for 64-bit applications.

As of StarLab 3.20, we no longer supply a component for 32-bit applications running on 64-bit machines. The ordinary 32-bit component will work with 32-bit applications on both 32-bit and 64-bit Windows.

3.2 Creating an Object

In order to use the library an object must be created. This object has methods which allow controlling the Ophir measuring devices. To create the object, the ProgID

`OphirLMMeasurement.CoLMMeasurement` or the GUID {B180613C-E514-4739-AADC-CAD4493910D7} can be used. How precisely to create a COM object depends on your development environment; see the provided samples for some possibilities.

Some environments require an interface name to be specified. For version 10.0 of the COM object (which comes with StarLab 3.50) the interface is `ICoLMMeasurement2`. The older interface `ICoLMMeasurement` will still work, but will not allow accessing the functions added in this version. Environments that do not require specifying the interface name will automatically use the newer interface.

3.3 Threading issues

The COM object provided is Apartment-threaded. This means (approximately) that it can be only used directly from the thread it was created in. The simplest way to use this library from multiple threads is to create a separate object in each thread. Since the state of the instruments is shared between objects, this has the same effect.

3.4 Error Codes

- 0x00000000 (S_OK): No Error
- 0x80004001 (E_NOTIMPL): Function Not Implemented
- 0x80070057 (E_INVALIDARG): Invalid Argument
- 0x80004005 (E_FAIL): Unspecified Failure
- 0x80040200: Device not opened
- 0x80040201: Device Already Opened
- 0x80040202: Drivers cannot be loaded
- 0x80040203: Load File Missing
- 0x80040300: Device Failed
- 0x80040301: Device Firmware is Incorrect
- 0x80040302: Sensor Failed
- 0x80040303: Sensor Firmware is Incorrect
- 0x80040304: Bad Device Handle
- 0x80040305: Bad Sensor Channel
- 0x80040306: This Sensor is Not Supported
- 0x80040307: Not Applicable in this Device
- 0x80040308: The Device is no longer Available
- 0x80040400: Save To Sensor Failed
- 0x80040401: Param Error
- 0x80040402: Failed to create Safe Array
- 0x80040403: Not Applicable in this Sensor
- 0x80040404: Value Out of Range
- 0x80040405: Command Failed
- 0x80040500: Stream Mode Not Started
- 0x80040501: A channel is in Stream Mode

The OphirLMMeasurement COM object follows the standard COM practice of returning an HRESULT from its methods. The HRESULT can be passed to [GetErrorFromCode](#) to get a descriptive string, or the standard COM `GetErrorInfo` function can be called (see documentation for your environment). Some client environments (such as VB6 and all .NET languages) do not return this HRESULT from the method call; instead they throw an exception when a method returns a failure HRESULT. This exception will normally contain within it the error number and the descriptive string.

3.5 Methods and Events

3.5.1 Device Communications

Close	CloseAll	GetKnownWirelessDevices
OpenUSBDevice	OpenWirelessDevice	ResetAllDevices
ResetDevice	ScanUSB	ScanWireless

3.5.2 General Information and Diagnostics

GetDeviceCalibrationDueDate	GetDeviceInfo	GetDriverVersion
GetErrorFromCode	GetSensorCalibrationDueDate	GetSensorInfo
GetVersion	IsSensorExists	

3.5.3 Sensor Configuration

AddWavelength	DeleteWavelength	GetDiffuser
GetFilter	GetLowFreqPowerPulseFreq	GetMeasurementMode
GetPulsedPowerPulseWidth	GetPulseLengths	GetRanges
GetThreshold	GetWavelengths	GetWavelengthsExtra
ModifyWavelength	SaveSettings	SetDiffuser
SetFilter	SetLowFreqPowerPulseFreq	SetMeasurementMode
SetPulsedPowerPulseWidth	SetPulseLength	SetRange
SetThreshold	SetWavelength	

3.5.4 Measurement Delivery

ConfigureStreamMode	GetData	StartStream
StopAllStreams	StopStream	

3.5.5 Trigger Settings

GetExtTrigModes	GetExtTrigOnOff	GetExtTrigWindowTime
SetExtTrigMode	SetExtTrigOnOff	SetExtTrigWindowTime

3.5.6 Events

DataReady	PlugAndPlay	
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3.5.7 ASCII Commands

Read	Write	
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3.5.8 Alphabetic Listing

Method	Communications	Diagnostics	Configuration	Measurement	Trigger	Legacy
AddWavelength			•			
Close	•					
CloseAll	•					
ConfigureStreamMode				•		
DataReady (event)						
DeleteWavelength			•			
GetData				•		
GetDeviceCalibrationDueDate		•				
GetDeviceInfo		•				
GetDiffuser			•			
GetDriverVersion		•				
GetErrorFromCode		•				
GetExtTrigModes					•	
GetExtTrigOnOff					•	
GetExtTrigWindowTime					•	
GetFilter			•			
GetKnownWirelessDevices	•					
GetLowFreqPowerPulseFreq			•			
GetMeasurementMode			•			
GetPulsedPowerPulseWidth			•			
GetPulseLengths			•			
GetRanges			•			
GetSensorCalibrationDueDate		•				
GetSensorInfo		•				
GetThreshold			•			
GetVersion		•				
GetWavelengths			•			
GetWavelengthsExtra			•			
IsSensorExists		•				
ModifyWavelength			•			
OpenUSBDevice	•					
OpenWirelessDevice	•					
PlugAndPlay (event)						
Read						•
ResetAllDevices	•					
ResetDevice	•					
SaveSettings			•			
ScanUSB	•					
ScanWireless	•					
SetDiffuser			•			
SetExtTrigMode					•	
SetExtTrigOnOff					•	
SetExtTrigWindowTime					•	
SetFilter			•			
SetLowFreqPowerPulseFreq			•			
SetMeasurementMode						
SetPulsedPowerPulseWidth			•			
SetPulseLength			•			
SetRange			•			
SetThreshold			•			
SetWavelength			•			
StartStream				•		
StopAllStreams				•		
StopStream				•		
Write						•

3.6 Detailed Explanations of Methods

3.6.1 Conventions

3.6.1.1 Sensor Identification

The set of devices that OphirLMMeasurement supports includes multichannel devices which may have more than one sensor attached. In order to provide a common interface across all devices, OphirLMMeasurement uniquely identifies a sensor based on the device that it is attached to (hDevice) and the channel to which it is attached. For devices with only one channel <channel> should be 0. For a 2-channel device it can be 0 or 1, and for a 4-channel device 0, 1, 2, or 3.

3.6.1.2 Strings

In order to guarantee compatibility with the various Development Environments, OphirLMMeasurement returns strings in the BSTR type. Arrays of strings are returned as VARIANT *.

3.6.1.3 Common Parameters

Parameter	Description
[in] LONG hDevice	A handle to an open device, received from OpenUSBDevice or OpenWirelessDevice .
[in] LONG channel	A channel number, 0 to 3, as applicable to the device.
[out] VARIANT* options	A list of values that a setting can take, in the form of an array of BSTR. If the setting is not applicable to the chosen sensor, the array will be empty.
[out] LONG* index	The 0-based index of the currently selected value of a setting. If the setting is not applicable to the chosen sensor, -1.
[in] LONG index	The 0-based index of a value of a setting. This must be in the range of the options valid for the setting, as returned in the <options> parameter of a Get... method.

3.6.2 Device Communications

These methods are responsible for managing the communications with Ophir devices. This includes finding and registering the device(s) within OphirLMMeasurement, starting a communications session, and closing down communications with the device(s).

3.6.2.1 Close

Name	Close
Parameters	[in] LONG hDevice
Use	This method ends the communication session with <hDevice>.
Return Codes	0x00000000 (S_OK): No Error 0x80040304: Bad Device Handle 0x80040501: A channel is in Stream Mode
See Also:	Device Communications , Alphabetic Listing , Common Parameters

3.6.2.2 CloseAll

Name	CloseAll
Parameters	None
Use	Equivalent to calling Close on all open devices.
Return Codes	0x00000000 (S_OK): No Error 0x80040501: A channel is in Stream Mode
See Also:	Device Communications , Alphabetic Listing

3.6.2.3 GetKnownWirelessDevices

Name	GetKnownWirelessDevices
Parameters	[out] VARIANT* serialNumbers
Use	This method is an infrastructure stub for future support and at present only returns an error code
Return Codes	0x80004001 (E_NOTIMPL): Function Not Implemented
See Also:	Device Communications , Alphabetic Listing

3.6.2.4 OpenUSBDevice

Name	OpenUSBDevice
Parameters	[in] BSTR serialNumber [out] LONG* hDevice
Use	Given <serialNumber> that was found with the ScanUSB method, will open a communication session with a device. <hDevice> is a handle which will be used for all communication with the device.
Return Codes	0x00000000 (S_OK): No Error 0x80070057 (E_INVALIDARG): Invalid Argument 0x80040201: Device Already Opened 0x80040300: Device Failed 0x80040301: Device Firmware is Incorrect 0x80040302: Sensor Failed 0x80040303: Sensor Firmware is Incorrect 0x80040306: This Sensor is Not Supported 0x80040308: The Device is no longer Available 0x80040405: Command Failed
See Also:	Device Communications , Alphabetic Listing , ScanUSB

3.6.2.5 OpenWirelessDevice

Name	OpenWirelessDevice
Parameters	• [in] BSTR name • [out] LONG* hDevice
Use	Given <name> that was found with the ScanWireless method, will open a communication session with a device. <hDevice> is a handle which will be used for all communication with the device.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80070057 (E_INVALIDARG): Invalid Argument • 0x80040201: Device Already Opened • 0x80040300: Device Failed • 0x80040301: Device Firmware is Incorrect • 0x80040302: Sensor Failed • 0x80040303: Sensor Firmware is Incorrect • 0x80040306: This Sensor is Not Supported • 0x80040308: The Device is no longer Available • 0x80040405: Command Failed
See Also:	Device Communications , Alphabetic Listing , ScanWireless

3.6.2.6 ResetAllDevices

Name	ResetAllDevices
Parameters	None
Use	Closes and restarts all open devices. After calling this function ScanUsb should be called to redetect the devices. Note: This function was only needed for older instruments. All current instruments detect automatically when the sensor is changed. Note: This method will only function when no devices are streaming data (i.e. any channels which were put into Stream mode with the StartStream method have been taken out of stream mode with StopStream or StopAllStreams).
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040501: A channel is in Stream Mode
See Also:	Device Communications , Alphabetic Listing

3.6.2.7 ResetDevice

Name	ResetDevice
Parameters	[in] LONG hDevice
Use	Restarts <hDevice>. Used after changing sensor attached to the device Note: This function was only needed for older instruments. All current instruments detect automatically when the sensor is changed. Note: This command method will only function when the device isn't streaming data (i.e. any channels which were put into Stream mode with the StartStream method have been taken out of stream mode with StopStream or StopAllStreams).
Return Codes	0x00000000 (S_OK): No Error 0x80040200: Device not opened 0x80040304: Bad Device Handle 0x80040308: The Device is no longer Available 0x80040405: Command Failed 0x80040501: A channel is in Stream Mode
See Also:	Device Communications , Alphabetic Listing , Common Parameters

3.6.2.8 ScanUSB

Name	ScanUSB
Parameters	[out] VARIANT* serialNumbers
Use	Scans the USB for all Ophir devices that are presently attached to the PC. Note: If the device is a Pulsar, this method will load it with its firmware. The firmware files are located under the directory where OphirLMMMeasurement.dll is located, in the subdirectory firmware. Updated firmware files, with the names FU4Axxx.hex, FU4Bxxx.hex, and FU4Fxxx.ttf, should be placed in the same location. <serialNumbers> will contain an array of BSTR.
Return Codes	0x00000000 (S_OK): No Error 0x80040202: Drivers cannot be loaded 0x80040203: Load File Missing 0x80040402: Failed to create Safe Array 0x80040501: A channel is in Stream Mode
See Also:	Device Communications , Alphabetic Listing , OpenUSBDevice

3.6.2.9 ScanWireless

Name	ScanWireless
Parameters	[out] VARIANT* names
Use	Scans for wireless Ophir devices. Currently the only wireless device supported is the Ariel. <names> will contain an array of BSTR.
Return Codes	0x00000000 (S_OK): No Error 0x80040202: Drivers cannot be loaded 0x80040203: Load File Missing 0x80040402: Failed to create Safe Array 0x80040501: A channel is in Stream Mode
See Also:	Device Communications , Alphabetic Listing , OpenWirelessDevice

3.6.3 General Information and Diagnostics

These methods provide general purpose information about the devices, sensors, and software that are presently attached and participating in the present communication session

3.6.3.1 GetDeviceInfo

Name	GetDeviceInfo
Parameters	• [in] LONG hDevice • [out] BSTR* deviceName • [out] BSTR* romVersion • [out] BSTR* serialNumber
Use	Gets name, ROM version and serial number of the device.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040308: The Device is no longer Available
See Also:	General Information and Diagnostics , Alphabetic Listing , Common Parameters

3.6.3.2 GetDeviceCalibrationDueDate

Name	GetDeviceCalibrationDueDate
Parameters	• [in] LONG hDevice • [out] DATE* dueDate
Use	Get the date this instrument is due to be calibrated. Note: Some instruments do not yet report the calibration due date and will return error 0x80040307.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040307: Not Applicable in this Device • 0x80040308: The Device is no longer Available
See Also:	General Information and Diagnostics , Alphabetic Listing , Common Parameters , GetSensorCalibrationDueDate

3.6.3.3 GetDriverVersion

Name	GetDriverVersion
Parameters	• [out] BSTR* info
Use	Get version of drivers that OphirLMMeasurement uses.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80004005 (E_FAIL): Unspecified Failure
See Also:	General Information and Diagnostics , Alphabetic Listing

3.6.3.4 GetErrorFromCode

Name	GetErrorFromCode
Parameters	• [in] LONG errorCode • [out] BSTR* info
Use	Given an error code, will pass back a string explanation of the error.
Return Codes	• 0x00000000 (S_OK): No Error
See Also:	General Information and Diagnostics , Alphabetic Listing

3.6.3.5 GetSensorCalibrationDueDate

Name	GetSensorCalibrationDueDate
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] DATE* dueDate
Use	Get the date this sensor is due to be calibrated. Note: Some instruments do not yet report the calibration due date and will return error 0x80040307.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040307: Not Applicable in this Device • 0x80040308: The Device is no longer Available
See Also:	General Information and Diagnostics , Alphabetic Listing , Common Parameters , GetDeviceCalibrationDueDate

3.6.3.6 GetSensorInfo

Name	GetSensorInfo
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] BSTR* serialNumber • [out] BSTR* sensorType • [out] BSTR* sensorName
Use	Gets serial number, type (thermopile, photodiode, pyroelectric), and name of sensor.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available
See Also:	General Information and Diagnostics , Alphabetic Listing , Common Parameters

3.6.3.7 GetVersion

Name	GetVersion
Parameters	• [out] LONG* version
Use	Get version of the OphirLMMeasurement COM object.
Return Codes	• 0x00000000 (S_OK): No Error
See Also:	General Information and Diagnostics , Alphabetic Listing

3.6.3.8 IsSensorExists

Name	IsSensorExists
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] VARIANT_BOOL* exists
Use	Check if a sensor is attached to <hDevice> <channel>.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040308: The Device is no longer Available
See Also:	General Information and Diagnostics , Alphabetic Listing , Common Parameters

3.6.4 Sensor Configuration

These methods are used to configure the measurement parameters of the sensor. Some of these methods apply to all types of sensors (e.g. Range methods) and some are sensor-type specific (e.g. Filter methods apply to Photodiode sensors only, Diffuser methods are only for Pyroelectric and PD Energy sensors, etc.)

3.6.4.1 AddWavelength

Name	AddWavelength
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG wavelength
Use	Add <wavelength> to list of favorite wavelengths of the selected sensor. Wavelength: Must be between the lower and upper wavelength limits as returned by the GetWavelengthsExtra method. Note: This method is only applicable for sensors with a continuous spectrum. The set of lasers cannot be altered in sensors with a discrete set of lasers settings.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor • 0x80040404: Wavelength Out of Range • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , DeleteWavelength , GetWavelengths , GetWavelengthsExtra , ModifyWavelength , SetWavelength

3.6.4.2 DeleteWavelength

Name	DeleteWavelength
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Delete wavelength of selected sensor specified in <index>. Note: If the currently selected wavelength is deleted, a different wavelength is automatically chosen. If there is only one wavelength left it cannot be deleted. Note: This method is only applicable for sensors with a continuous spectrum. The set of lasers cannot be altered in sensors with a discrete set of lasers settings
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , AddWavelength , GetWavelengths , GetWavelengthsExtra , ModifyWavelength , SetWavelength

3.6.4.3 GetDiffuser

Name	GetDiffuser
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	Get the diffuser state of the selected sensor and what the diffuser options are. For sensors that are not equipped with an adjustable diffuser, <index> will always return 0. This method applies to Pyroelectric and PD Energy sensors only.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetDiffuser

3.6.4.4 GetFilter

Name	GetFilter
Parameters	• in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	Get the filter state of the selected sensor and what the filter options are. For sensors that are not equipped with an adjustable filter, <index> will always return 0. This method applies to Photodiode sensors only.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetFilter

3.6.4.5 GetLowFreqPowerPulseFreq

Name	GetLowFreqPowerPulseFreq
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] DOUBLE* value • [out] DOUBLE* min • [out] DOUBLE* max
Use	Get the current pulse frequency for low frequency power measurement mode, and the maximum and minimum supported values. This method applies to Photodiode sensors only, and is only supported on some instruments.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetLowFreqPowerPulseFreq

3.6.4.6 GetMeasurementMode

Name	GetMeasurementMode
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	Get the measurement mode of the selected sensor and what the measurement modes are. This method applies to all types of sensors.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetMeasurementMode

3.6.4.7 GetPulsedPowerPulseWidth

Name	GetPulsedPowerPulseWidth
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* value • [out] LONG* min • [out] LONG* max
Use	Get the current pulse width for pulsed power measurement mode, and the maximum and minimum supported values, in milliseconds. This method applies to Thermopile sensors only, and is only supported on some instruments.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetPulsedPowerPulseWidth

3.6.4.8 GetPulseLengths

Name	GetPulseLengths
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	Get the selected pulse length of the sensor in selected channel and what the pulse length options are. This method applies to Pyroelectric and PD Energy sensors only. The strings returned are the maximum pulse length (in time) for each of the various pulse length settings.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetPulseLength

3.6.4.9 GetRanges

Name	GetRanges
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	Get the measurement range of the selected sensor and the list of all available measurement ranges. This method applies to all types of sensors.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetRange

3.6.4.10 GetThreshold

Name	GetThreshold
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	Get the selected threshold setting of the sensor in selected channel and what the threshold options are. This method applies to Thermopile sensors measuring energy and to some Pyroelectric sensors as well. For Thermopile sensors, the set of available thresholds will generally be "LOW MEDIUM HIGH". For Pyroelectric sensors with adjustable thresholds, there is a threshold setting per pulse length.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , SetThreshold

3.6.4.11 GetWavelengths

Name	GetWavelengths
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	Gets the selected wavelength of selected sensor and list of all available wavelengths. This method applies to all types of sensors. Response is the same format for sensors with discrete and continuous spectrums. For additional spectrum information use GetWavelengthsExtra. For sensors with a continuous spectrum, these strings will be numeric, for sensors with discrete spectrums, these strings may contain non-numeric characters as well (e.g. VIS or NIR).
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , AddWavelength , DeleteWavelength , GetWavelengthsExtra , ModifyWavelength , SetWavelength

3.6.4.12 GetWavelengthsExtra

Name	GetWavelengthsExtra
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] VARIANT_BOOL*modifiable • [out] LONG* minWavelength • [out] LONG* maxWavelength
Use	Get additional wavelength information of selected channel. If spectrum is continuous, then • modifiable: True • minWavelength, maxWavelength: minimum and maximum limits for wavelength adjustment with the AddWavelength and ModifyWavelength methods If spectrum is discrete then • modifiable: False • minWavelength, maxWavelength: Not applicable
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , AddWavelength , DeleteWavelength , GetWavelengths , ModifyWavelength , SetWavelength

3.6.4.13 ModifyWavelength

Name	ModifyWavelength
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index • [in] LONG wavelength
Use	Modify selected sensor's favorite wavelength stored at <index> to the value <wavelength>. Note: <wavelength> must be between the lower and upper wavelength limits as returned by the GetWavelengthsExtra method. Note: This method is only applicable for sensors with a continuous spectrum. The set of lasers cannot be altered in sensors with a discrete set of lasers settings.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040404: Wavelength Out of Range • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , AddWavelength , DeleteWavelength , GetWavelengths , GetWavelengthsExtra , SetWavelength

3.6.4.14 SaveSettings

Name	SaveSettings
Parameters	• [in] LONG hDevice • [in] LONG channel
Use	Save all configuration parameters of selected sensor.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040400: Save To Sensor Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters

3.6.4.15 SetDiffuser

Name	SetDiffuser
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Set diffuser state of selected sensor to <index> This method applies to Pyroelectric and PD Energy sensors only.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , GetDiffuser

3.6.4.16 SetFilter

Name	SetFilter
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Set filter state of selected sensor to <index>. This method applies to Photodiode sensors only.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , GetFilter

3.6.4.17 SetLowFreqPowerPulseFreq

Name	SetLowFreqPowerPulseFreq
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] DOUBLE value
Use	Set the pulse frequency for low frequency power measurement mode. This method applies to Photodiode sensors only, and is only supported on some instruments.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor • 0x80040404: Value Out of Range • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , GetLowFreqPowerPulseFreq

3.6.4.18 SetMeasurementMode

Name	SetMeasurementMode
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Set measurement mode of selected sensor to <index>. This method applies to all types of sensors.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also	Sensor Configuration , Alphabetic Listing , Common Parameters , GetMeasurementMode

3.6.4.19 SetPulsedPowerPulseWidth

Name	SetPulseLength
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG value
Use	Get the pulse width for pulsed power measurement mode, in milliseconds. This method applies to Thermopile sensors only, and is only supported on some instruments.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor • 0x80040404: Value Out of Range • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , GetPulsedPowerPulseWidth

3.6.4.20 SetPulseLength

Name	SetPulseLength
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Set pulse length setting of the selected sensor to <index>. This method applies to Pyroelectric and PD Energy sensors only.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , GetPulseLengths

3.6.4.21 SetRange

Name	SetRange
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Set measurement range of selected sensor to <index>. This method applies to all types of sensors.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , GetRanges

3.6.4.22 SetThreshold

Name	SetThreshold
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Set threshold of selected sensor to <index>. This method applies to Thermopile sensors measuring energy and to some Pyroelectric sensors as well.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , GetThreshold

3.6.4.23 SetWavelength

Name	SetWavelength
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG index
Use	Set wavelength of selected sensor to <index>. This method applies to all types of sensors.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Sensor Configuration , Alphabetic Listing , Common Parameters , AddWavelength , DeleteWavelength , GetWavelengths , GetWavelengthsExtra , ModifyWavelength

3.6.5 Measurement Delivery

See section [Data Streams](#) for detailed explanations of the various streams of data that come out of the devices and through OphirLMMeasurement, setting up the stream, starting and stopping it, and parsing the response.

3.6.5.1 ConfigureStreamMode

Name	ConfigureStreamMode
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG mode • [in] LONG nValue
Use	Configure special measurement delivery options for the selected device and channel. See section Data Streams for detailed explanations of setting up the stream, starting and stopping it, and parsing the response.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Measurement Delivery , Alphabetic Listing , Common Parameters

3.6.5.2 GetData

Name	GetData
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] VARIANT* arrayValue // array of double • [out] VARIANT* arrayTimestamp // array of double • [out] VARIANT* arrayStatus // array of LONG
Use	This method retrieves the buffered data that has been received from the meter. It should be called after the DataReady event has been received, or at appropriate intervals. The returned data is removed from the buffer. If called when no new data is available, the arrays will be empty. The three arrays are the same size, with corresponding elements containing the value, timestamp and status for a single item of data. Note that not every item of data represents a measurement, therefore it is essential to check the status to know what is being returned. See section Data Streams for detailed explanations of setting up the stream, starting and stopping it, and interpreting the response.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array • 0x80040500: Stream Mode Not Started
See Also:	Measurement Delivery , Alphabetic Listing , Common Parameters , Data Streams , GetData Status Codes

3.6.5.3 StartStream

Name	StartStream
Parameters	• [in] LONG hDevice • [in] LONG channel
Use	Instruct the selected sensor to begin data streaming measurements to the PC. When StartStream is called, the COM object flushes the buffer of previously gathered measurements. OphirLMMMeasurement will fire the DataReady event when a new data packet has arrived and is ready for reading by the client application. See section Data Streams for detailed explanations of setting up the stream, starting and stopping it, and parsing the response.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80004005 (E_FAIL): Unspecified Failure • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Measurement Delivery , Alphabetic Listing , Common Parameters , Data Streams

3.6.5.4 StopAllStreams

Name	StopAllStreams
Parameters	None
Use	Stops data streams of all sensors. See section Data Streams for detailed explanations of setting up the stream, starting and stopping it, and parsing the response.
Return Codes	0x00000000 (S_OK): No Error
See Also:	Measurement Delivery , Alphabetic Listing , Data Streams

3.6.5.5 StopStream

Name	StopStream
Parameters	[in] LONG hDevice [in] LONG channel
Use	Stops data streaming from selected sensor. See section Data Streams for detailed explanations of setting up the stream, starting and stopping it, and parsing the response.
Return Codes	0x00000000 (S_OK): No Error 0x80040200: Device not opened 0x80040304: Bad Device Handle 0x80040305: Bad Sensor Channel 0x80040308: The Device is no longer Available 0x80040500: Stream Mode Not Started
See Also:	Measurement Delivery , Alphabetic Listing , Common Parameters , Data Streams

3.6.6 Trigger Settings

These methods are for driving the Pulsar's and Centauri's External Trigger mechanisms. External Trigger functionality is provided on the Pulsar with Pyroelectric and PD Energy sensors, and on the Centauri with all sensors.

See section [External Trigger Settings](#) for a brief explanation of the External Trigger mechanism. For a more detailed explanation please see **External Triggers and Missing Pulses** in the *StarLab_User_Manual* that is included in the StarLab Installation package.

3.6.6.1 GetExtTrigModes

Name	GetExtTrigModes
Parameters	[in] LONG hDevice [out] LONG* index [out] VARIANT* options
Use	Gets the selected external trigger mode of the device and the set of available external trigger modes.
Return Codes	0x00000000 (S_OK): No Error 0x80040200: Device not opened 0x80040304: Bad Device Handle 0x80040307: Not Applicable in this Device 0x80040308: The Device is no longer Available 0x80040402: Failed to create Safe Array
See Also:	Trigger Settings , Alphabetic Listing , Common Parameters , External Trigger Modes

3.6.6.2 GetExtTrigOnOff

Name	GetExtTrigOnOff
Parameters	• [in] LONG hDevice • [in] LONG channel • [out] LONG* index • [out] VARIANT* options
Use	To get the external trigger state.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040402: Failed to create Safe Array
See Also:	Trigger Settings , Alphabetic Listing , Common Parameters

3.6.6.3 GetExtTrigWindowTime

Name	GetExtTrigWindowTime
Parameters	• [in] LONG hDevice • [out] LONG* extTrigWindow
Use	Get the external trigger window time of <hDevice> in microseconds. Window time can be between 1 and 65535 uS on the Pulsar, and between 1 and 50000 uS on the Centauri.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040307: Not Applicable in this Device • 0x80040308: The Device is no longer Available
See Also:	Trigger Settings , Alphabetic Listing , Common Parameters , External Trigger Window

3.6.6.4 SetExtTrigMode

Name	SetExtTrigMode
Parameters	• [in] LONG hDevice • [in] LONG index
Use	Set the external trigger mode of <hDevice>.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040307: Not Applicable in this Device • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040405: Command Failed
See Also:	Trigger Settings , Alphabetic Listing , Common Parameters , External Trigger Modes

3.6.6.5 SetExtTrigOnOff

Name	SetExtTrigOnOff
Parameters	• [in] LONG hDevice • [in] LONG channel • [in] LONG* index
Use	Set the external trigger state.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040305: Bad Sensor Channel • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040403: Not Applicable in this Sensor • 0x80040405: Command Failed • 0x80040501: A channel is in Stream Mode
See Also:	Trigger Settings , Alphabetic Listing , Common Parameters

3.6.6.6 SetExtTrigWindowTime

Name	SetExtTrigWindowTime
Parameters	• [in] LONG hDevice • [in] LONG extTrigWindowTime
Use	Set the external trigger window time of <hDevice> in microseconds. Window time can be between 1 and 65535 uS on the Pulsar, and between 1 and 50000 uS on the Centauri.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040307: Not Applicable in this Device • 0x80040308: The Device is no longer Available • 0x80040401: Param Error • 0x80040405: Command Failed
See Also:	Trigger Settings , Alphabetic Listing , Common Parameters , External Trigger Window

3.6.7 Events

3.6.7.1 DataReady

Name	DataReady
Parameters	• [in] LONG hDevice • [in] LONG channel
Use	This event is fired to inform the client application that the sensor at <hDevice> <channel> has delivered a new data payload. OphirLMMeasurement will not deliver the data to the client application until it calls the GetData method. Note: Further DataReady events for the specified sensor will not be fired until OphirLMMeasurement's flag has been cleared by the client application calling GetData.
Return Codes	N/A
See Also:	Measurement Delivery , Events , Alphabetic Listing , Common Parameters

3.6.7.2 PlugAndPlay

Name	PlugAndPlay
Parameters	None
Use	Fired when a device that has been found by the ScanUSB method has been disconnected from the system. The client application can rescan the USB to discover which device has been disconnected.
Return Codes	N/A
See Also:	Device Communications , General Information and Diagnostics , Events , Alphabetic Listing

3.6.8 ASCII Commands

We recommend using the OphirLMMMeasurement methods wherever possible. Features that do not have a direct method call can be accessed using the ASCII command methods. These methods may also be useful when porting from RS232 based solutions.

The function sequence is always a call to Write followed by a call to Read, i.e. every Write must be followed by a Read, and every Read must be preceded by a Write. Failure to adhere to this sequence may cause communication trouble.

For a description of available commands and device compatibility, please see “**Ophir User Commands.pdf**” included in the installation.

3.6.8.1 Read

Name	Read
Parameters	• [in] LONG hDevice • [out] BSTR* reply
Use	Read <reply> from <hDevice>. <reply> is in the form of an ASCII string.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040308: The Device is no longer Available • 0x80040405: Command Failed
See Also:	ASCII Commands , Alphabetic Listing , Common Parameters , Working with Legacy Methods

3.6.8.2 Write

Name	Write
Parameters	• [in] LONG hDevice • [in] BSTR command
Use	Write <command> to <hDevice>. <command> is in the form of an ASCII string.
Return Codes	• 0x00000000 (S_OK): No Error • 0x80040200: Device not opened • 0x80040304: Bad Device Handle • 0x80040308: The Device is no longer Available • 0x80040405: Command Failed
See Also:	ASCII Commands , Alphabetic Listing , Common Parameters , Working with Legacy Methods

4 Data Streams

This section describes how measurement data is delivered from Ophir instruments to a client application through the OphirLMMeasurement COM object, and how a client application typically works with that data.

The methods that control data delivery are listed in [Measurement Delivery](#). This section focuses on how those methods are used together.

4.1 Stream Formats

OphirLMMeasurement provides three types of data stream formats from the various devices to the client application.

- Standard
- Turbo
- Immediate

4.1.1 **Standard**

This is the default data stream that OphirLMMeasurement uses. Data is buffered within OphirLMMeasurement and which then fires the [DataReady](#) event to inform the client application that new data is available. This event can be fired as often as once every 50 milliseconds depending on how quickly the data is coming in and how quickly the client application calls the [GetData](#) method to read out the data from OphirLMMeasurement.

Besides OphirLMMeasurement's buffering, the Juno and Pulsar devices have hardware buffering mechanisms that enable them to reach very high measurement frequencies (10 KHz and 25 KHz respectively)

4.1.2 **Turbo**

The Vega and Nova-II devices can deliver measurements at 130Hz in Standard format. To deliver every pulse at higher frequencies (2000Hz) they need to use in Turbo Mode. Using the [ConfigureStreamMode](#) method, the client application can tell OphirLMMeasurement that it wants to work in Turbo Mode and what is the expected frequency of the laser being measured.

To set	mode	nValue
Turbo mode	0	0: off; 1: on
Turbo frequency	1	Any frequency acceptable to the device

Turbo Mode is not relevant for other devices.

4.1.3 **Immediate**

For clients that want to synchronize their Ophir measurements with something else (for example energy of a pulse together with a beam profile of the pulse captured by a camera in another piece of software such as BeamGage), OphirLMMeasurement provides Immediate Mode streams. Using the [ConfigureStreamMode](#) method to put OphirLMMeasurement in Immediate Mode, whenever a new measurement arrives, OphirLMMeasurement immediately fires the [DataReady](#) event instead of buffering the reading. Do not use immediate mode with high frequency data as the data delivery mechanism will not be able to keep up. This feature is supported by Juno and Pulsar devices. Note that on the Pulsar all channels are affected simultaneously.

To set	mode	nValue
Immediate mode	2	0: off; 1: on

4.2 Measurement Streaming Model

Measurement data is delivered using a streaming model. Once streaming is started, the instrument continuously sends data to the COM object, which buffers it until the client application retrieves it.

A typical measurement session follows this sequence:

1. Open the device and configure the sensor.
2. Start streaming measurements.
3. Retrieve buffered data during or after the measurement session.
4. Stop streaming and close the device.

All sensor types and measurement modes use the same data retrieval mechanism and data format.

4.3 Setup and Start

Before starting a measurement stream, the client application must:

1. **Detect and open a device**
Use [ScanUSB](#) to find connected instruments, then call [OpenUSBDevice](#) to open one of them and obtain a device handle (hDevice).
2. **Optionally configure the sensor**
Sensor parameters such as measurement mode, range, wavelength, filter, diffuser, pulse length, or threshold may be configured before streaming begins. See [Sensor Configuration](#). The configuration methods generally come in pairs, a Get method which returns a list of options and the currently selected option, and a Set method which allows choosing a new option. Configuration methods cannot be called while a channel is streaming.
3. **Start streaming**
Call [StartStream](#)(hDevice, channel) to begin measurement delivery.
When streaming starts, any previously buffered data is discarded.

4.4 Data Collection

Once streaming has started, the COM object buffers incoming data from the instrument. How and when the data is retrieved is up to the client application.

Valid data collection strategies include:

- Responding to the [DataReady](#) event and retrieving data as it arrives.
- Periodically polling at a fixed interval.
- Allowing data to accumulate in the buffer and retrieving it once at the end of the measurement session.
- Clearing the buffer by calling [GetData](#) (and discarding the results), performing measurements, then calling [GetData](#) to retrieve the measurements.

4.4.1 Retrieving Data

Measurement data is retrieved by calling [GetData](#)(hDevice, channel, arrayValue, arrayTimestamp, arrayStatus)

This method returns three parallel arrays of equal length:

- **Value** – the measurement value (e.g. Watts or Joules) as a double
- **Timestamp** – time of measurement, in milliseconds as a double
- **Status** – status code describing the measurement as a LONG

Each index across the arrays corresponds to one data item. After [GetData](#) returns, the retrieved items are removed from the internal buffer. If no new data is available, all arrays are empty.

4.4.2 Interpreting Status Codes

Not every data item represents a valid measurement. The status code must always be checked to determine the meaning of each returned value.

Status codes indicate conditions such as:

- Valid measurement
- Overrange or saturation
- Pulse acquisition states
- Beam tracking components
- Informational notifications (e.g. temperature or filter state changes)

The complete list is provided in [GetData Status Codes](#).

4.5 Stopping and Closing

When measurements are complete, stop streaming using [StopStream](#) or [StopAllStreams](#), then close the device using [Close](#). After a device is closed, no further configuration or measurement calls should be made for that handle.

4.6 GetData Status Codes

Each data item returned by [GetData](#) includes a status code describing its meaning. Client applications must interpret this code before using the measurement value. The following is the list of status codes that can be returned. Other codes that may be reported are reserved for internal use or future expansion.

Status Code	Meaning	When and Where
0	OK	Power/Energy/Exposure measurements for all sensors
1	Overrange	Power and Energy measurements for all sensors
2	Saturated	Power and Energy measurements for all sensors
3	Missing	Energy with Pyroelectric and PD Energy sensors on a Pulsar or Centauri device
4	Reset state	Single shot energy or Pulsed Power with a Thermopile sensor
5	Waiting	Single shot energy or Pulsed Power with a Thermopile sensor
6	Summing	Single shot energy or Pulsed Power with a Thermopile sensor
7	Timeout	Single shot energy or Pulsed Power with a Thermopile sensor
8	Peak Over	Single shot energy or Pulsed Power with a Thermopile sensor
9	Energy Over	Single shot energy or Pulsed Power with a Thermopile sensor
0x010000	X measurement OK	Track measurements with a BeamTrack sensor
0x010001	X measurement Error	Track measurements with a BeamTrack sensor
0x020000	Y measurement OK	Track measurements with a BeamTrack sensor
0x020001	Y measurement Error	Track measurements with a BeamTrack sensor
0x030000	Size measurement OK	Track measurements with a BeamTrack sensor
0x030001	Size measurement Error	Track measurements with a BeamTrack sensor
0x030002	Size measurement Warning	Track measurements with a BeamTrack sensor
0x040001	Filter State Change	Photodiode sensors with built-in filter state detection reporting a change in filter state (in to out or vice-versa)
0x050000	Pulse frequency	Power and Energy with Pyroelectric and PD Energy sensors
0x100000	Temperature	Specific sensors
0x200000	Alert Hot	Specific sensors
0x300000	Pulse Width	Specific sensors
0x400000	PfP Energy	Specific sensors

5 External Trigger Settings

The Pulsar and Centauri devices have a sophisticated External Trigger mechanism. This allows synchronization between pulse measurements and other inputs (or outputs) in the measurement environment.

Note: On the Pulsar, External Trigger functionality (including Missing Pulses) is available for Pyroelectric and PD Energy sensors only. On the Centauri, edge triggers are available for Pyroelectric and PD Energy sensors only; level triggers are available for all sensors.

Typical setup of the External Trigger methods is as follows

1. Set Trigger Mode
2. Set Window (for rising and falling edge modes)
3. Set External Trigger ON for the relevant sensors

5.1 External Trigger Modes

The Pulsar External Trigger can be set to one of 6 modes (4 input modes, 2 output modes). The Centauri External Trigger can be set to one of 4 input modes – the output modes are not supported.

<i>Input Modes (Centauri and Pulsar)</i>	<i>Description</i>
Rising Edge	The device is sensitive to a trigger on the RISING EDGE of the input. The trigger is valid for a pulse arriving during a window of time after the active (rising) edge The inactive (falling) edge of the signal is ignored. Missing Pulses are recorded when an External Trigger edge is received, but no pulse arrives within the Window Time. Pulses are ignored if they arrive outside the Window Time.
Falling Edge	The device is sensitive to a trigger on the FALLING EDGE of the input. The trigger is valid for a pulse arriving during a window of time after the active (falling) edge The inactive (rising) edge of the signal is ignored. Missing Pulses are recorded when an External Trigger edge is received, but no pulse arrives within the Window Time. Pulses are ignored if they arrive outside the Window Time.
High Level	Pulses are recorded only when the input signal is at a HIGH LEVEL. Any pulse arriving while the signal is high is counted. Any pulse arriving while the signal is low is ignored. No Missing Pulses are recorded in this mode.
Low Level	Pulses are recorded only when the input signal is at a LOW LEVEL. Any pulse arriving while the signal is low is counted. Any pulse arriving while the signal is high is ignored. No Missing Pulses are recorded in this mode.

<i>Output Modes (Pulsar only)</i>	<i>Description</i>
Active High	Every time a pulse arrives on the sensor detector the output goes high for 10us and then returns back to low. The default level of the output with no pulses is low.
Active Low	Every time a pulse arrives on the sensor detector the output goes low for 10us and then returns back to high. The default level of the output with no pulses is high.

5.2 External Trigger Window

For trigger modes that define edge-triggered measurements (rising or falling), this defines the window of time before and after the edge that a pulse is considered to be in synch with the trigger. If there is no pulse during this window, then the device will report a MISSING MEASUREMENT.

